

REMOTE CONTROLLED WATER CLEANING BOAT

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Abstract: Water plays an essential role in our daily lives and is vital for sustaining life. Many towns and communities rely on rivers and other natural water bodies as major sources of drinking water. Unfortunately, the increasing amount of waste found in these waters is not just an inconvenience it poses serious dangers to the environment, to us, and to our loved ones. Even a single piece of litter thrown on the ground can eventually end up in streams and rivers, contributing to pollution. Therefore, it is necessary to remove this waste efficiently. The Water Cleaning Boat Project aims to clean rivers and lakes by Remote controller, with the goal of reducing pollution while minimizing human effort. In this project, a special boat is designed, built, and tested. It uses a conveyor belt driven by a DC motor to pick up floating garbage like plastic, leaves, and other waste from the surface of the water. The boat's electrical system is controlled by a Motor Driver and ESP32. which can be operated wirelessly through a mobile phone. This smart method helps keep water bodies clean and supports a healthier environment for aquatic plants and animals. This project focuses not only on technological innovation but also on creating a sustainable environment through the effective use of science, technology, and community support.

Keywords: Water Cleaning Boat, ESP32, Motor Driver, Mobile Phone, Aquatic plants, Sustainable Environment, Technology.

INTRODUCTION

As we all are aware about 71 percent of the Earth is covered with water, and it has always been extremely important for human life. From ancient times, people have lived near rivers and lakes because they help us survive and grow. Even today, we depend on water bodies for many needs. However, with increasing industrialization, we have started harming these natural resources. Wastewater often contains materials that do not decompose. These non-biodegradable wastes are dangerous because they can cause climate changes, floods, and pollution. To protect our environment, it is our responsibility to save and maintain these resources. For many years, people have dumped waste into water bodies, causing heavy pollution. Now it is time to clean up the mess we created usually, people clean rivers and lakes manually by collecting trash, plastic, and other waste from the surface of the water. But this method is risky, expensive, and takes a lot of time and labor. This is why we need a smarter and more eco-friendly system. One such solution is a water cleaning boat. This machine can remove waste from water bodies with very little human effort. It collects floating waste from lakes, rivers, and ponds. The machine works using a conveyor belt powered by a battery. The belt rotates and picks up waste from the water, carrying it into a collection bin. The boat moves based on the directions given through a mobile application. A camera on the boat helps the operator choose the best direction for cleaning. In simple words, it is a mobile application based water cleaning boat that picks up trash from the water using a moving belt.

LITERATURE REVIEW

Literature review on research from years (2021-2025) around water cleaning boat that is controlled by mobile application is given below:

In study 1, River Cleaning Boat (2025) proposed by Sneha Gupta and her team suggested a system demonstrate that the robot has a Bluetooth device that allows it to communicate with humans. A smartphone application known as Bluetooth electronics allows a person to operate a robot that is within Bluetooth range.

In study 2, Water Cleaning RC Boat (2024) proposed by Archana S.K and her team members there main goal of the project was the development of an RC (Remote Control) water boat that offers solution to deal with water pollution in smaller water bodies such as ponds, lakes and rivers. The boat is equipped with sensors, a collector bin and a mobile application interface for efficient operation in various water conditions.

In study 3, Design and Testing of Remote Control Trash Collector For Lake Water (2023) proposed by Rakshit C.P, Prajwal B.Y. Many prior works have explored floating boats or robots to remove surface waste from lakes, rivers, and ponds often using conveyor-belt or chain-drive mechanisms mounted on a floating platform. For instance, a remote-monitored aqua garbage collecting robot used a belt mechanism plus sensors and remote control to gather floating plastic bottles and debris from water surfaces. Similarly, several designs employ a motor-driven conveyor and storage bin on a small boat or floating hull to lift trash out of water and collect it onboard.

In study 4, A Fully Autonomous Water Surface Cleaning Robot with A Novel Coverage Path Planning Method(2022), proposed by Jiannan Zhu and team designs a fully autonomous water surface cleaning robot, SMURF, which achieves high efficiency water surface cleaning without human operation. In addition, they propose a novel coverage path planning the water surface method and an

improved model predictive controller. The real-world experiment shows that SMURF works well in different kinds of water bodies and achieves much higher efficiency than traditional water surface cleaning methods.

In study 5, IoT Based Water Surface Cleaning and Quality Checking Boat (2021) proposed by B. Saran Raj, S. Murli and team there work aims to create a garbage gathering object to monitor the lakes and ponds and also measure the condition of water by measuring several parameters. The main feature of there project is to collect of floating objects from the surface of the water into a trash container. The Secondary work of there project was to measure the water quality by using sensors, by measuring the water parameters we can predict the unfavourable conditions of water living creatures, pH sensor and turbidity sensor are used in our project to measure the acidity and basicity of the water and turbidity level of the water respectively. The main motive was to clean the water surface and prevent the aquatic life.

PROBLEM STATEMENT

Various water bodies like lakes, rivers, and ponds are increasingly polluted due to floating waste, including plastics, vegetation, and other solid debris. The manual cleaning methods are labor intensive, extremely time consuming, and often unsafe for workers who must navigate contaminated or difficult to access areas. The solutions that are existing are quite expensive, lack remote operability, or require a specialized training. Hence, there is a need for an affordable, user friendly, and application controlled system that can effectively collect floating waste from water surfaces. A water cleaning boat based on a mobile application can enable real time monitoring, remote navigation, and efficient waste collection, it also reduces human efforts ensuring the human safety and less consumption of time.

OBJECTIVES OF THE PROJECT

The primary objectives of the water cleaning boat are as follows:

- 1.The primary objective is to design and implement a mobile application based water cleaning boat that collects floating waste from water bodies using a mobile application for control and monitoring.
- 2.To minimize manual labor and operational cost for water cleaning by using automation and mobile application based boat.
- 3.Create a low-cost and scalable prototype that can be improved and deployed in larger bodies of water. Improve efficiency and safety by reducing the human contact with the polluted water.
- 4.Reduce water pollution by removing plastics, leaves, and other floating waste from lakes, rivers, and ponds which will help in providing a clean water to the aquatic life as well as the humans.
5. To improve water quality by removing pollutants like plastics, chemicals, and debris, thereby protecting aquatic life and human health, reducing disease risks (like typhoid), and promoting sustainable environments.

SYSTEM MODEL AND COMPONENTS ANALYSIS

The proposed system is a small boat that floats on water and can move in any direction to collect waste materials from the surface. It acts as a working prototype of a water cleaning device that gathers floating debris and stores it in a collection bin automatically. The boat can be programmed, increased in size if needed, and operated from a distance. The power supply enables the boat to carry out its functions, while a motor driver controls the speed and direction of two motors. These motors run and the propellers that move the boat. They are operated using a mobile app called RC Bluetooth Controller, which connects to the boat via Bluetooth. In the app, the control keys are: F indicates forward, B indicates backward, R indicates right, and L indicates left. There are also Start and Stop buttons to begin or stop the boat's motion.

Table 1: List of Components

Sr. No	Component	Specification	Quantity
1	DC Motor	12V	4
2	Motor Driver	L298N	1
3	Microcontroller	ESP32	1
4	Conveyor Belt	-	1
5	Battery	12V	1
6	Garbage Collector	Plastic Container	1
7	Mobile Application	Bluetooth RC Controller	1

Motor Driver: The L298N Motor Driver Module is used to operate DC and stepper motors. It contains an L298N IC and a built- in 5V regulator that needs a 5V power supply. This module can run up to four DC motors, or two DC motors with both speed and direction control. In this system, three 12V DC geared motors with 60 RPM and a maximum load current of 350mA are used to rotate the boat's propellers and move the conveyor belt. These motors are connected to the OUT1, OUT2, OUT3, and OUT4 pins of the motor driver to control their movement.

Microcontroller ESP32: Here ESP32 act as a center unit of the project. The ESP32 can connect with other components to offer Wi-Fi and Bluetooth features. It receives wireless commands from a smartphone app or a remote controller through Bluetooth or Wi-Fi. These commands allow the operator to control how the boat moves and which direction it goes.

DC Motor: In a water cleaning boat, DC motors act as the primary operational muscles. They provide the mechanical energy needed to propel the boat, navigate it, power mechanical arms or paddles that pull in floating debris, and drive the conveyor belts that lift waste into an onboard collection bin.

Conveyor Belt: The conveyor belt uses the motor's torque to drive a chain mechanism that lifts waste from the water's surface and drops it into a storage bin called the garbage collector. A polyvinyl conveyor belt is used for this process. The belt's movement is controlled by system through a motor driver circuit. PVC pipes are placed underneath the boat to support the air-tube guiding mechanism. This setup directs floating debris toward the boat, where the conveyor belt then gathers and moves the waste into the collector.

Lithium Battery: A lithium-ion battery is used as the main power source for water-cleaning boat. The lithium battery provides DC power to gear motors, which drive the propellers for moving the boat forward/backward. The conveyor belt mechanism collects floating waste. Because lithium batteries give a stable voltage, the motors run smoothly and efficiently. The user removes the SD card and inserts it into a laptop/PC. The stored scan data is now ready for processing.

Mobile Application: The Mobile app used in this particular project is Bluetooth RC Controller which is connected to ESP32 mobile app for a Bluetooth RC controller, you need to program the ESP32 with code that communicates via Bluetooth, and use a mobile app like an existing "RC controller" application.

METHODOLOGY

The process begins with system initialization and proceeds through the following stages- Start and Initialization, in this stage the system is turned on, and all input/output ports and necessary hardware are initialized. Command Receiving Loop, in this is loop the system waits for the command we can say that it waits for the input signal from the smart phone. If a Bluetooth command is not received i.e "No", the system continues to wait for a signal. If a command is received "Yes", the system proceeds to the next step. Execution of Task, the water cleaning boat moves or performs an action as directed by the received command the command received can be move left/right or forward/backward. Checking the new task, after completion of current action, the system checks if there is another task pending. If there is a new task "Yes", the process loops back to receive further input signals/commands from the smartphone. If there are new plastic and leaves the conveyor belt lifts the waste from the water surface and drops it into a collection tray

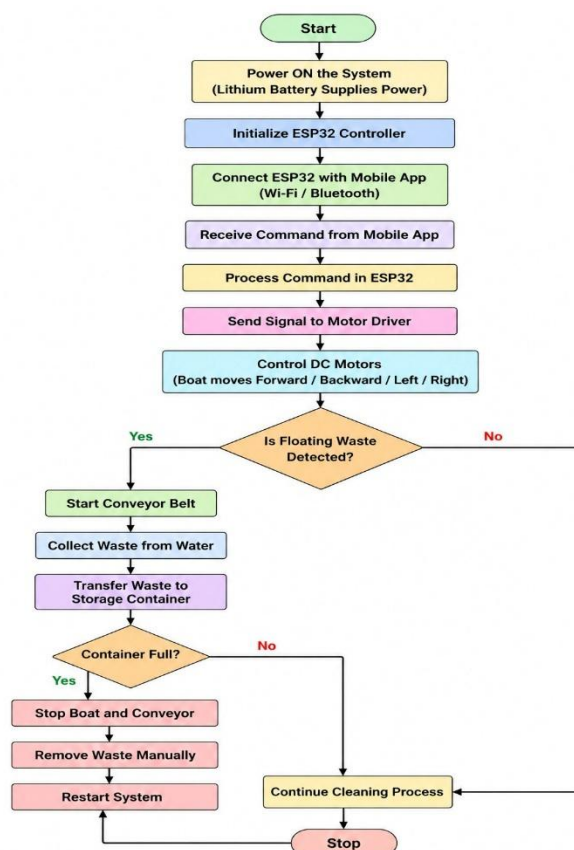


Fig 1: Flow Chart

HARDWARE IMPLEMENTATION

The hardware implementation of this project consist of an ESP32, Motor Driver and Batteries. The ESP32 controls two DC motors using an L298N motor driver. The ESP32 sends signals to the motor using some specific pins. The enA and enB pins of the motor driver are respectively connected to the pins 5 and 23 of the ESP32, these control how fast each motor spins. The direction of the motors is controlled using four pins that are IN1, IN2, IN3, and IN4, which are connected to ESP32 pins 22, 21, 19, and 18. The two motors are attached to the driver outputs Motor A is connected to OUT1–OUT2 and Motor B is connected to OUT3–OUT4. A separate AA battery pack gives power to the motors by connecting to the main power input of the motor driver. The motor driver also provides a 5V output, which is used to power the ESP32 through its VIN pin. All grounds (ESP32, motor driver, and battery) are connected together so everything works smoothly and signals remains steady.

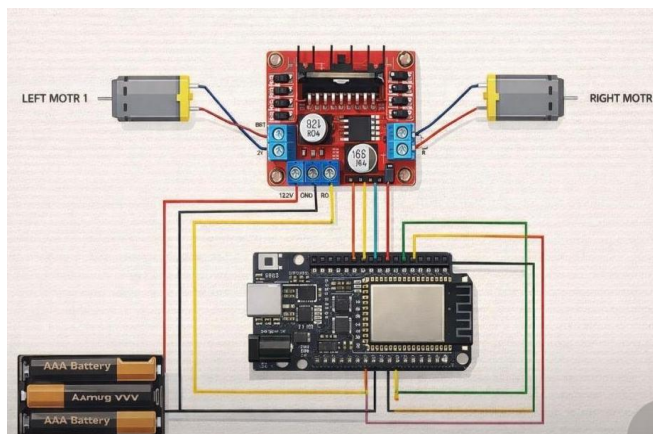


Fig 2: Hardware Implementation

SOFTWARE IMPLEMENTATION

A bluetooth RC control mobile app is used in this respective project, it wirelessly navigates the water cleaning boat and manage its operation. The primary function of this application is to control the boats movement remotely such as forward, backward, left or right. After pairing the phone with the ESP32 through Bluetooth, the app sends simple commands whenever buttons like forward, backward, left, or right are pressed. The ESP32 receives these commands and moves the motors accordingly. The app is easy to install and use, and it doesn't require any special setup. This makes controlling the car smooth and convenient for our entire team during testing and demonstration. The app also provides a user-friendly interface with virtual buttons or joysticks, making the boat easy to operate, even for non-experts. This approach also provides a cost-effective, user-friendly, and efficient solution for cleaning small to medium sized water bodies without manual contact with the water.

APPLICATIONS

- 1) Cleaning rivers, lakes, and ponds: Removes floating trash, plastic, leaves, and debris from surface of the water in order to make the water clean.
- 2) Reducing water pollution as water pollution is increasing it helps prevent harmful substances from accumulating in water and in keeping aquatic life safe.
- 3) Monitoring water quality some boats can be equipped with sensors to track pollution levels, and the data can be viewed on the mobile application
- 4) Smart and remote operation allows users to control the boat or schedule cleaning sessions directly from a smartphone without going inside the water and working for several hours.
- 5) Urban and recreational areas useful for maintaining clean water in city parks, tourist spots, and public lakes.
- 6) Educational and environmental awareness demonstrates sustainable practices and encourages people to protect water bodies.
- 7) Cost and labor efficiency it is cost effective and labor aren't required one can easily handle it with the help of mobile application providing human safety

RESULTS

The use of a water cleaning boat based on mobile application has shown a very positive results. It effectively removes trash, leaves, and other floating debris from rivers, lakes, and ponds, making the water clean and safe for the aquatic life. It not only reduces pollution but also helps to protect the environment and maintain better water quality. The mobile application allows real-time monitoring, showing areas that have been cleaned and the amount of waste collected, which makes the cleaning process more efficient. Additionally, the automated operation saves time and reduces the need for manual labor. Overall, this technology provides an eco-friendly solution for maintaining clean water bodies and also being the effective step in order to decrease the water pollution.



Fig 3: Results

Table 2: Result Performance

Sr No.	Condition	Output
1	Power Supply is ON	ESP32 and Motor Driver start functioning
2	ESP32 receives command to move forward	Boat moves front
3	ESP32 receives command to move Backward	Boat moves Back
4	ESP32 receives command to move left	Left motor slows and boat turns left
5	ESP32 receives command to move right	Right motor slows and boat turns right
6	Conveyor belt is activated	Conveyor belt start rotating
7	Floating waste reaches conveyor belt	Waste is lifted from water

CONCLUSION

Water is one of our most valuable natural resources, but it is increasingly affected by pollution and floating waste. This debris blocks sunlight from reaching underwater plants, disrupts aquatic life, and reduces overall water quality. To address this issue, the proposed system uses a conveyor mechanism to collect floating waste and transfer it into a storage container. The boat is controlled wirelessly, allowing smooth operation of both the conveyor and movement systems for effective waste collection. By providing a simple, efficient, and cost-effective method for cleaning water surfaces, this system helps protect aquatic ecosystems and contributes to a cleaner and more sustainable environment.

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